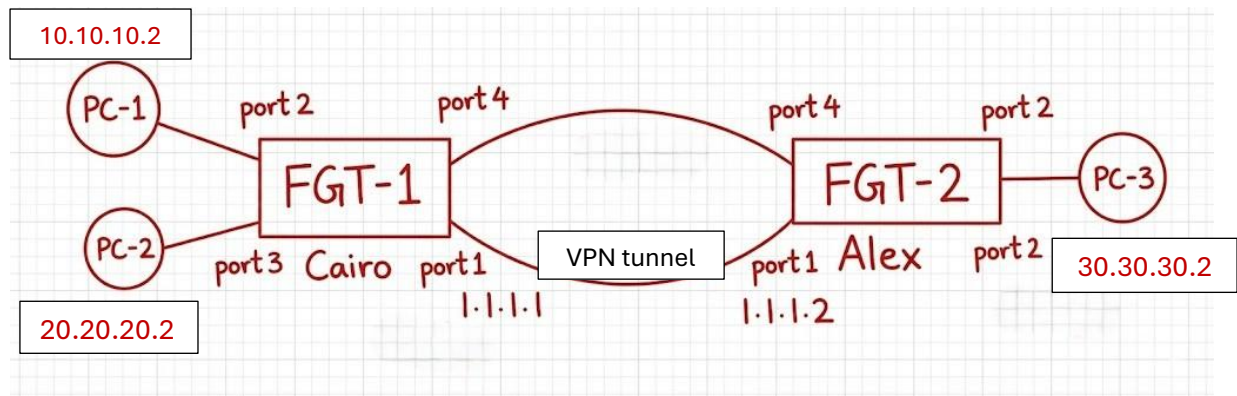


Week 2: IPsec VPN Configuration

Task: Establish an IPsec VPN tunnel between two FortiGate devices.

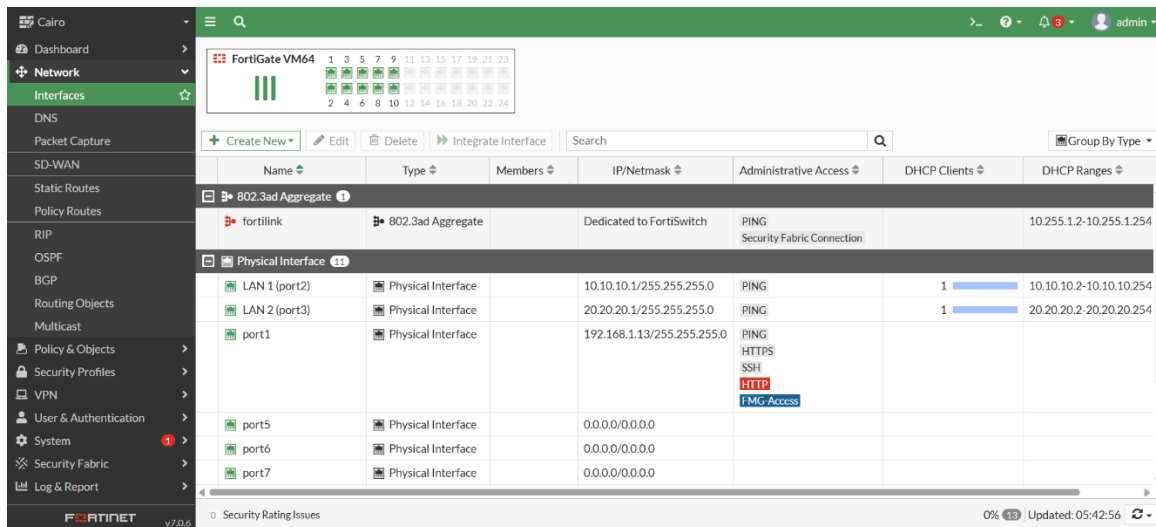
We established this topology:



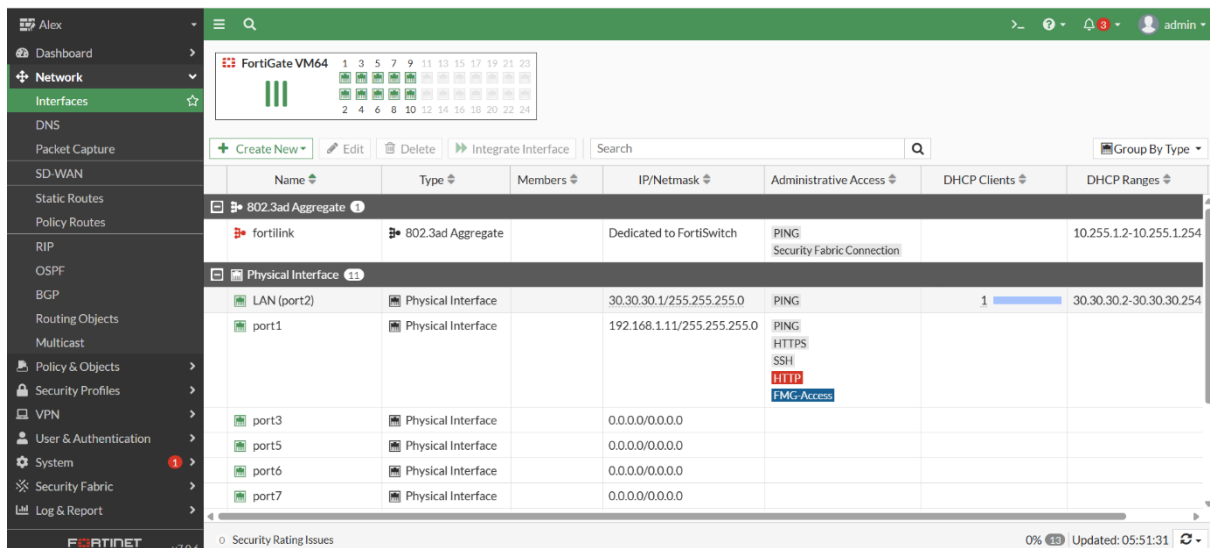
Configuration:

1. Configure the interfaces and make port 1 is the management port

Cairo:



Alex:



2. Configure the vpn tunnel between the two firewalls and choose traffic selector from the network 10.10.10.0 to 30.30.30.0 in Cairo firewall and reverse in the Alex firewall

Cairo:

The screenshot shows the FortiGate Cairo firewall configuration interface for an IPsec VPN tunnel. The left sidebar contains the navigation menu with options like Dashboard, Network, Policy & Objects, Security Profiles, VPN, and User & Authentication. The main panel is titled 'Edit VPN Tunnel' and shows the configuration for a tunnel named 'Cairo to Alex'.

Network Section:

- IP Version: IPv4
- Remote Gateway: Static IP Address
- IP Address: 1.1.1.2
- Interface: WAN (port4)
- Local Gateway: ☒ Primary IP 1.1.1.1
- Mode Config: ☐
- NAT Traversal: ☒ Enable
- Keepalive Frequency: 10
- Dead Peer Detection: ☒ Disable
- DPD retry count: 3
- DPD retry interval: 20 s
- Forward Error Correction: ☐ Egress ☐ Ingress

Authentication Section:

- Method: Pre-shared Key
- Pre-shared Key: 123456
- IKE Version: 1
- Mode: ☒ Aggressive ☐ Main (ID protection)

Phase 1 Proposal Section:

- Encryption: DES
- Authentication: MD5
- Diffie-Hellman Groups: ☐ 32 ☐ 31 ☐ 30 ☐ 29 ☐ 28 ☐ 27 ☐ 21 ☐ 20 ☐ 19 ☐ 18 ☐ 17 ☐ 16 ☒ 15 ☒ 14 ☒ 5 ☒ 2 ☐ 1
- Key Lifetime (seconds): 86400
- Local ID:

XAUTH Section:

- Type: Disabled

Alex

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

192.168.1.110

FortiGate v7.0.6

Edit VPN Tunnel

NameAlex to Cairo

CommentsComments

Network

Remote Gateway : Static IP Address (1.1.1.1) , Local Gateway: 1.1.1.2, Interface : port4

Authentication

MethodPre-shared Key

Pre-shared Key123456

IKE

Version12

ModeAggressiveMain (ID protection)

Phase 1 Proposal

Algorithms : DES-MD5

Diffie-Hellman Groups : 14, 5, 2

XAUTH

Type : Disabled

Additional Information

API Preview

References

IPsec VPNs

Guides

IPsec VPN Cookbook Recipes

VPN Setup on FortiClient

Configuring an IPsec VPN Connection

Documentation

Online Help

Video Tutorials

OKCancel

Phase 1 Proposal

Add

EncryptionDES

AuthenticationMD5

Diffie-Hellman Groups

Key Lifetime (seconds)86400

Local ID

XAUTH

TypeDisabled

Alex

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

192.168.1.110

FortiGate v7.0.6

Edit VPN Tunnel

Edit Phase 2

Name30 - 10

CommentsComments

Local AddressSubnet30.30.30.0/255.255.255

Remote AddressSubnet10.10.10.0/255.255.255

Advanced...

Phase 2 Proposal

Add

EncryptionDES

AuthenticationMD5

EncryptionDES

AuthenticationSHA1

Enable Replay Detection

Enable Perfect Forward Secrecy (PFS)

Local PortAll

Remote PortAll

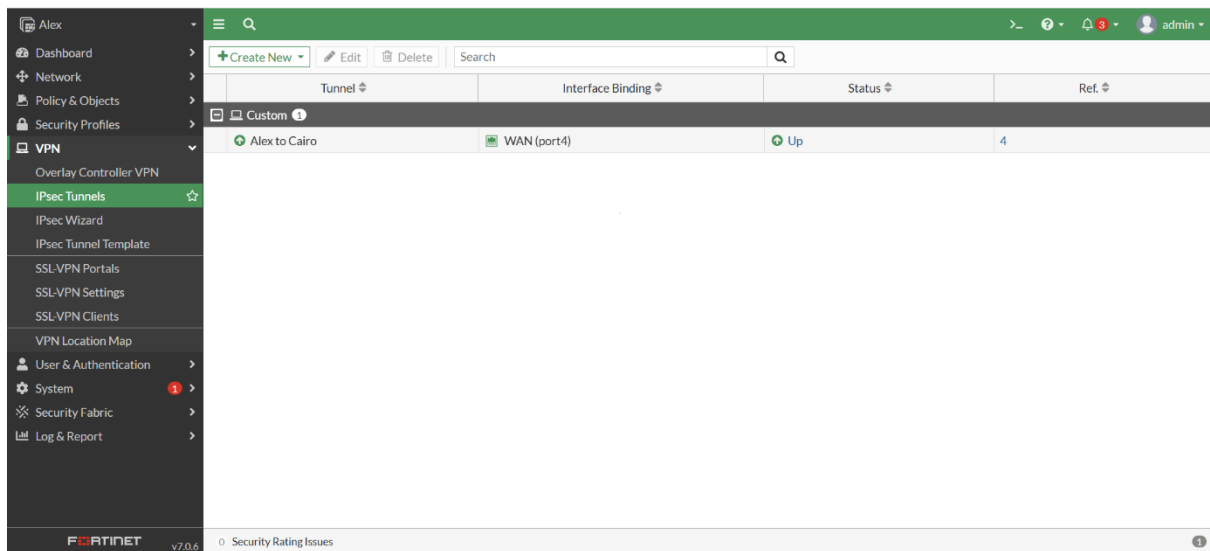
ProtocolAll

Auto-negotiate

Autokey Keep Alive

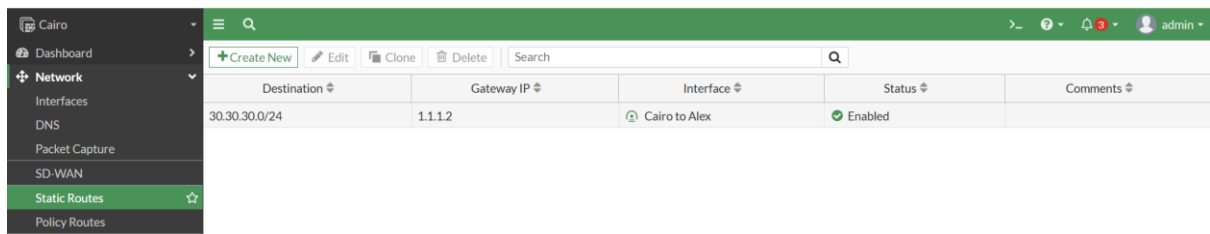
Key LifetimeSeconds

Seconds43200

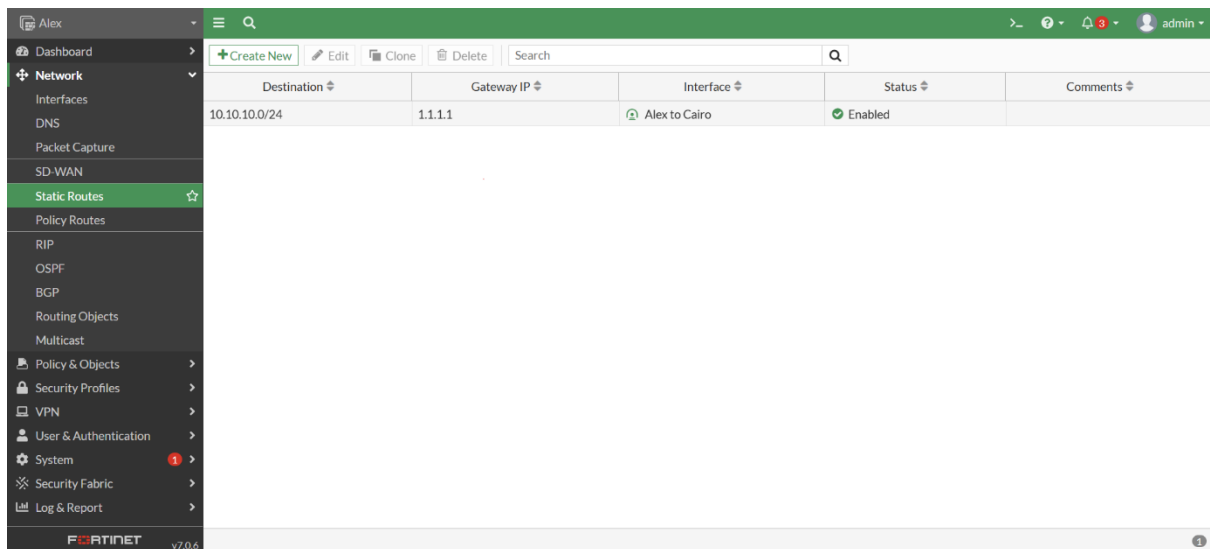


3. Configure the static route

Cairo:

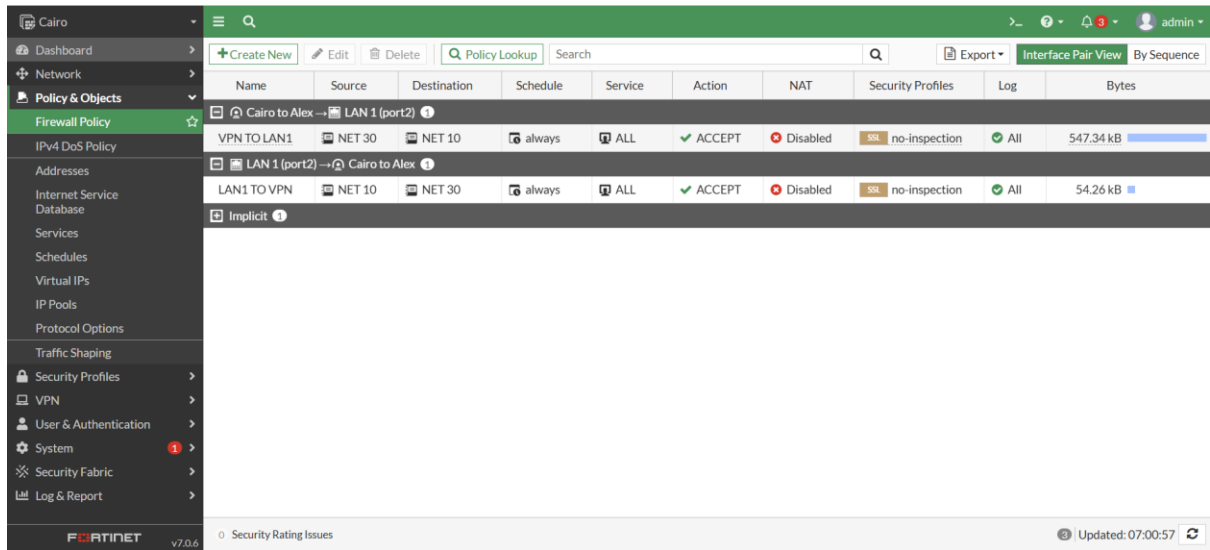


Alex:



4. Configure the policy between network 10.10.10.0 and 30.30.30.0

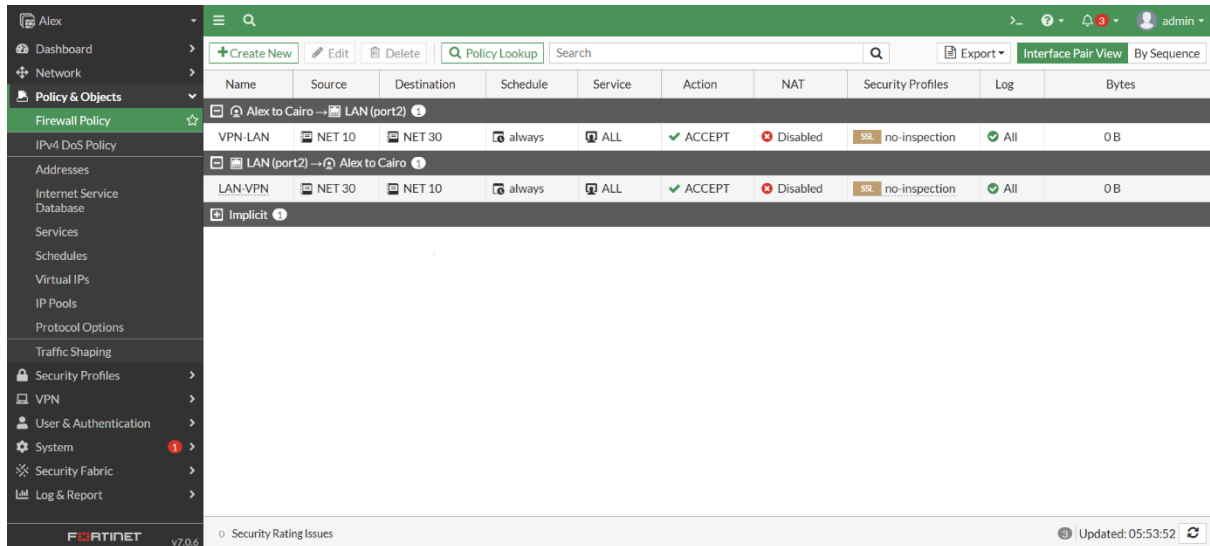
Cairo:



The screenshot shows the FortiGate Cairo interface for configuring Firewall Policies. The left sidebar lists various configuration sections, with 'Policy & Objects' selected. The main area displays a table of policies. The first policy, 'Cairo to Alex -> LAN 1 (port2)', is selected and its details are shown below the table. The policy is configured for 'VPN TO LAN1' with source 'NET 30' and destination 'NET 10'. It is set to 'always' schedule, 'ALL' service, 'ACCEPT' action, 'Disabled' NAT, 'no-inspection' security profile, and 'All' log. The policy is currently disabled and has a size of 547.34 kB.

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes	
Cairo to Alex -> LAN 1 (port2)	VPN TO LAN1	NET 30	NET 10	always	ALL	ACCEPT	Disabled	no-inspection	All	547.34 kB
LAN 1 (port2) -> Cairo to Alex	LAN 1 (port2)	NET 10	NET 30	always	ALL	ACCEPT	Disabled	no-inspection	All	54.26 kB
Implicit										

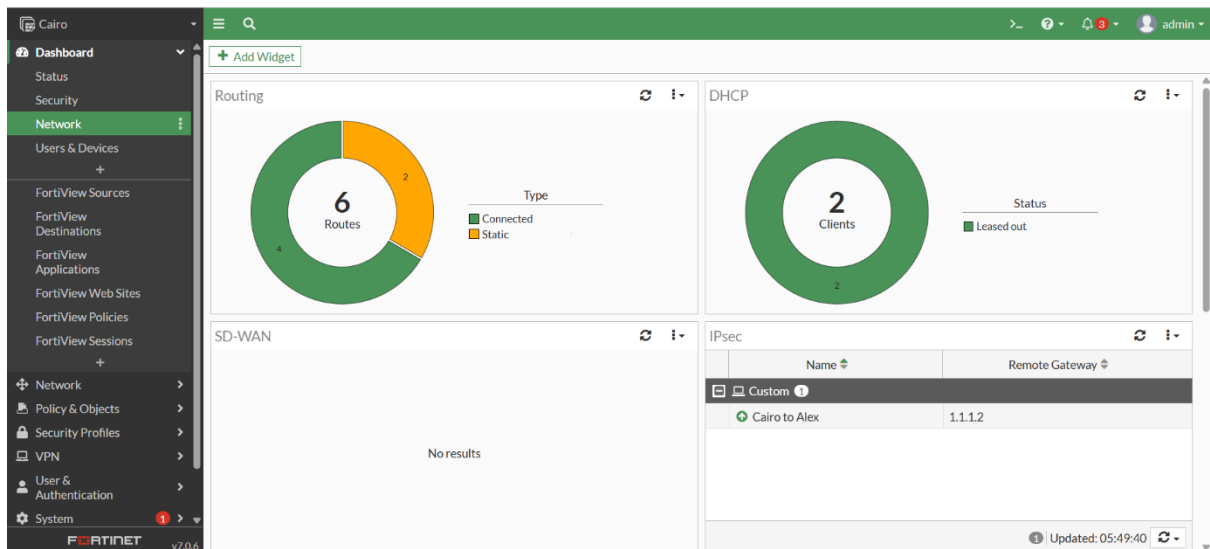
Alex:



The screenshot shows the FortiGate Alex interface for configuring Firewall Policies. The left sidebar lists various configuration sections, with 'Policy & Objects' selected. The main area displays a table of policies. The first policy, 'Alex to Cairo -> LAN (port2)', is selected and its details are shown below the table. The policy is configured for 'VPN-LAN' with source 'NET 10' and destination 'NET 30'. It is set to 'always' schedule, 'ALL' service, 'ACCEPT' action, 'Disabled' NAT, 'no-inspection' security profile, and 'All' log. The policy is currently disabled and has a size of 0 B.

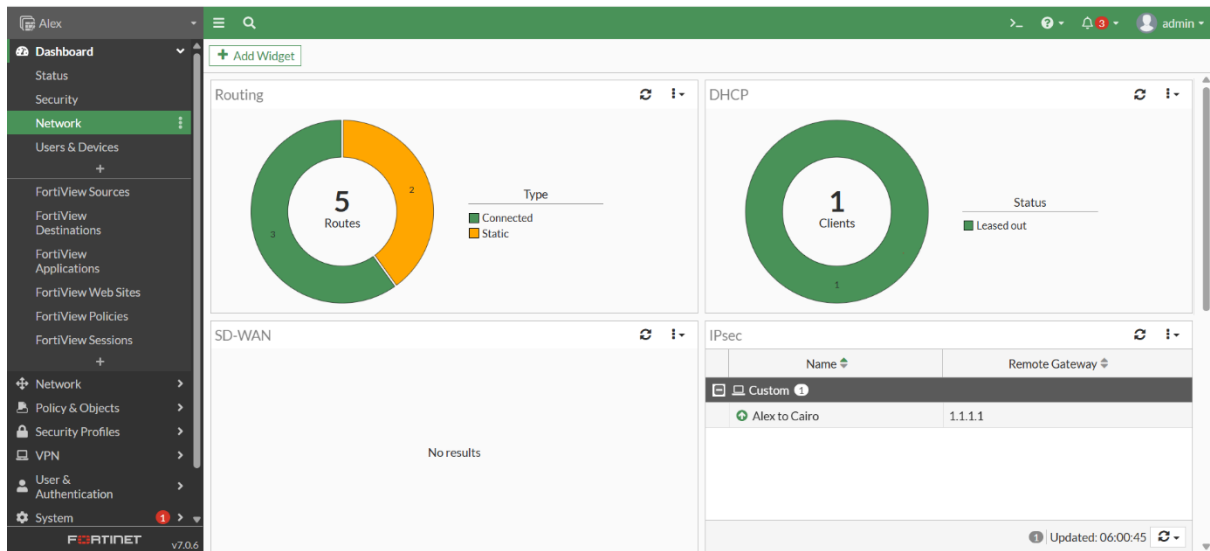
Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes	
Alex to Cairo -> LAN (port2)	VPN-LAN	NET 10	NET 30	always	ALL	ACCEPT	Disabled	no-inspection	All	0 B
LAN (port2) -> Alex to Cairo	LAN-VPN	NET 30	NET 10	always	ALL	ACCEPT	Disabled	no-inspection	All	0 B
Implicit										

So the Vpn tunnel now is up and running



The screenshot shows the FortiGate Cairo Dashboard. The left sidebar lists various configuration sections, with 'Network' selected. The main area displays four widgets: 'Routing' showing 6 routes (4 connected, 2 static), 'DHCP' showing 2 clients (2 leased out), 'SD-WAN' showing no results, and 'IPsec' showing a single tunnel 'Cairo to Alex' with remote gateway '1.1.1.2'.

Name	Remote Gateway
Custom	
Cairo to Alex	1.1.1.2



Test Results:

So let us see the result when we ping from PC-1 to PC-3

```
root@slitaz:~# ping 30.30.30.2
PING 30.30.30.2 (30.30.30.2): 56 data bytes
64 bytes from 30.30.30.2: seq=0 ttl=62 time=0.382 ms
64 bytes from 30.30.30.2: seq=1 ttl=62 time=0.642 ms
64 bytes from 30.30.30.2: seq=2 ttl=62 time=1.180 ms
64 bytes from 30.30.30.2: seq=3 ttl=62 time=1.170 ms
64 bytes from 30.30.30.2: seq=4 ttl=62 time=1.086 ms
64 bytes from 30.30.30.2: seq=5 ttl=62 time=2.421 ms
64 bytes from 30.30.30.2: seq=6 ttl=62 time=2.337 ms
64 bytes from 30.30.30.2: seq=7 ttl=62 time=2.245 ms
64 bytes from 30.30.30.2: seq=8 ttl=62 time=2.245 ms
64 bytes from 30.30.30.2: seq=9 ttl=62 time=0.443 ms
64 bytes from 30.30.30.2: seq=10 ttl=62 time=0.848 ms
64 bytes from 30.30.30.2: seq=11 ttl=62 time=2.359 ms
64 bytes from 30.30.30.2: seq=12 ttl=62 time=2.237 ms
64 bytes from 30.30.30.2: seq=13 ttl=62 time=1.622 ms
64 bytes from 30.30.30.2: seq=14 ttl=62 time=0.570 ms
64 bytes from 30.30.30.2: seq=15 ttl=62 time=0.996 ms
64 bytes from 30.30.30.2: seq=16 ttl=62 time=1.479 ms
```

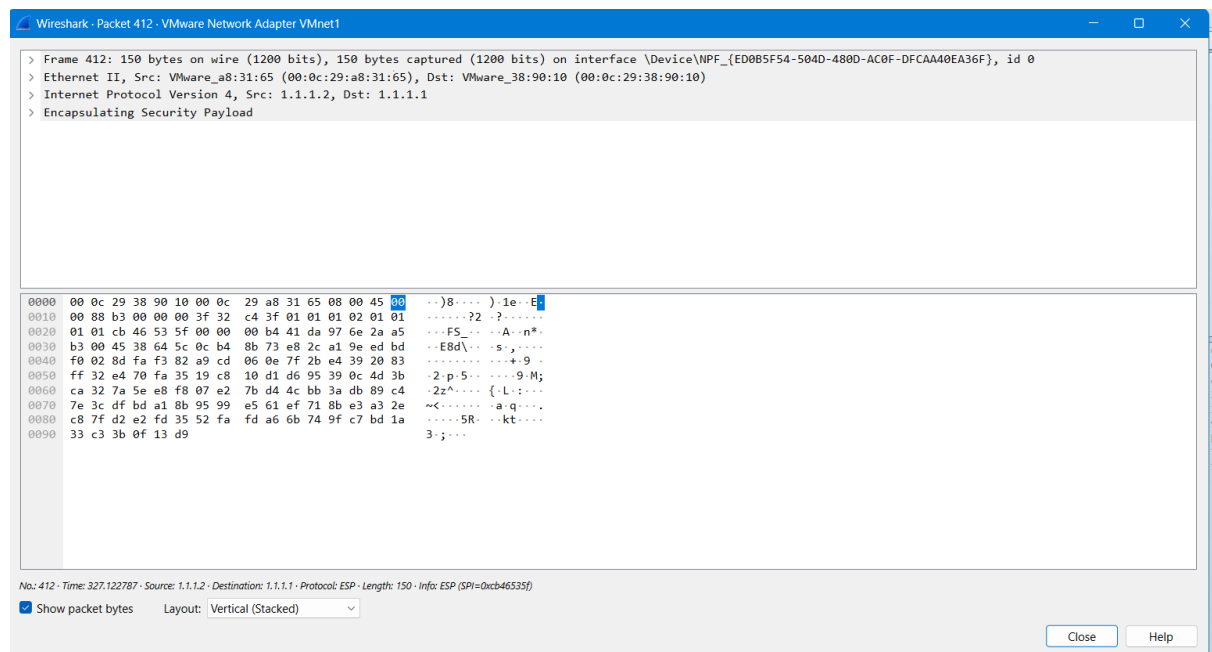
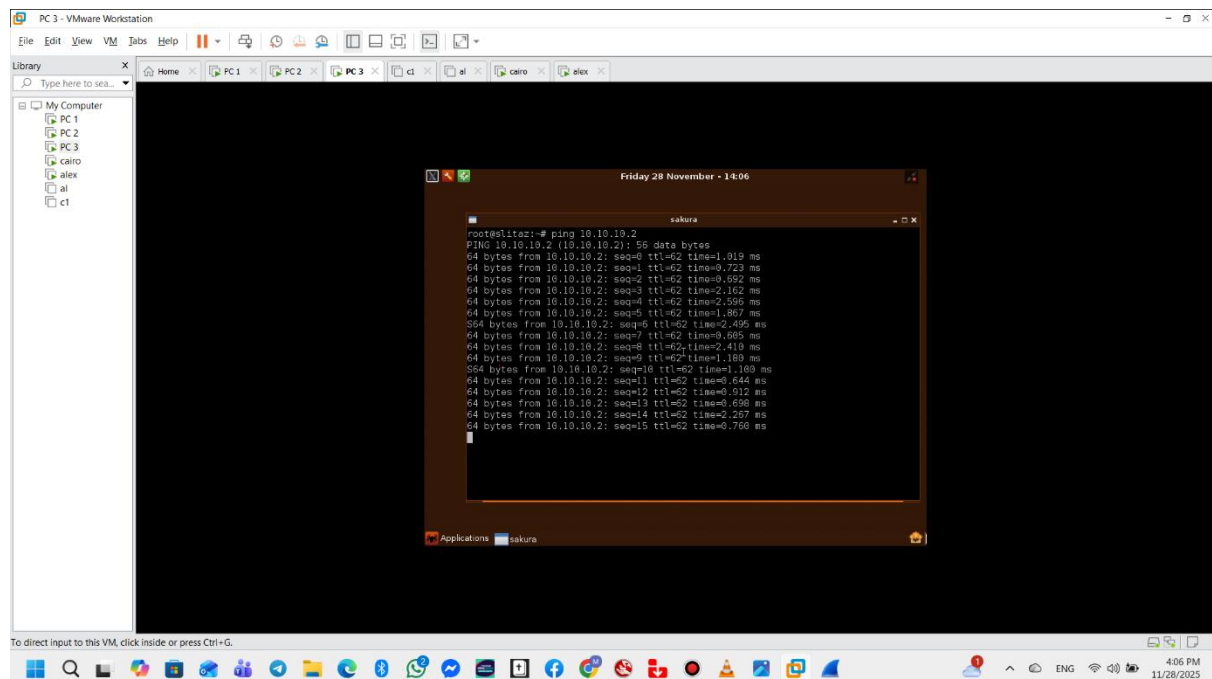
Capturing from VMware Network Adapter VMnet1

No.	Time	Source	Destination	Protocol	Length	Info
126	239.977887	1.1.1.2	1.1.1.1	ESP	150	ESP (SPI=0xc64653f)
127	240.978456	1.1.1.1	1.1.1.2	ESP	150	ESP (SPI=0xd68f6695)
128	240.978971	1.1.1.2	1.1.1.1	ESP	150	ESP (SPI=0xc64653f)
129	241.686319	192.168.74.1	224.0.0.251	NDNS	234	Standard query response 0x0000 PTR DESKTOP-33ESL6P._dosvc._tcp.local SRV 0 6800 DESKTOP-33ESL6P.local TXT
130	241.688916	fe80::be11:ca39:421::ff02::fb	224.0.0.251	NDNS	254	Standard query response 0x0000 PTR DESKTOP-33ESL6P._dosvc._tcp.local SRV 0 6800 DESKTOP-33ESL6P.local TXT
131	241.611082	192.168.74.1	224.0.0.251	NDNS	93	Standard query 0x0000 ANY DESKTOP-33ESL6P._dosvc._tcp.local, "QM" question
132	241.612204	fe80::be11:ca39:421::ff02::fb	224.0.0.251	NDNS	113	Standard query 0x0000 ANY DESKTOP-33ESL6P._dosvc._tcp.local, "QM" question
133	241.670397	192.168.74.1	224.0.0.251	NDNS	93	Standard query 0x0000 ANY DESKTOP-33ESL6P._dosvc._tcp.local, "QM" question
134	241.674147	fe80::be11:ca39:421::ff02::fb	224.0.0.251	NDNS	113	Standard query 0x0000 ANY DESKTOP-33ESL6P._dosvc._tcp.local, "QM" question
135	241.980323	1.1.1.1	1.1.1.2	ESP	150	ESP (SPI=0xd68f6695)
136	241.980629	1.1.1.2	1.1.1.1	ESP	150	ESP (SPI=0xc64653f)
137	242.132460	192.168.74.1	224.0.0.251	NDNS	93	Standard query 0x0000 ANY DESKTOP-33ESL6P._dosvc._tcp.local, "QM" question
138	242.133523	fe80::be11:ca39:421::ff02::fb	224.0.0.251	NDNS	113	Standard query 0x0000 ANY DESKTOP-33ESL6P._dosvc._tcp.local, "QM" question
139	242.394488	192.168.74.1	224.0.0.251	NDNS	299	Standard query response 0x0000 PTR, cache flush DESKTOP-33ESL6P._dosvc._tcp.local SRV, cache flush 0 6800 DESKTOP-33ESL6P.local T...

Frame 1: 234 bytes on wire (1872 bits), 234 bytes captured (1872 bits) on interface \Device\NPF_{E0B85F54-504D-480B-AC0F-DFCAA4EA36F}, id 0
> Ethernet II, Src: VMware_c0:00:01:00:50:56:c0:00:01, Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
> Internet Protocol Version 4, Src: 192.168.74.1, Dst: 224.0.0.251
> User Datagram Protocol, Src Port: 5353, Dst Port: 5353
> Multicast Domain Name System (response)

Packets: 139

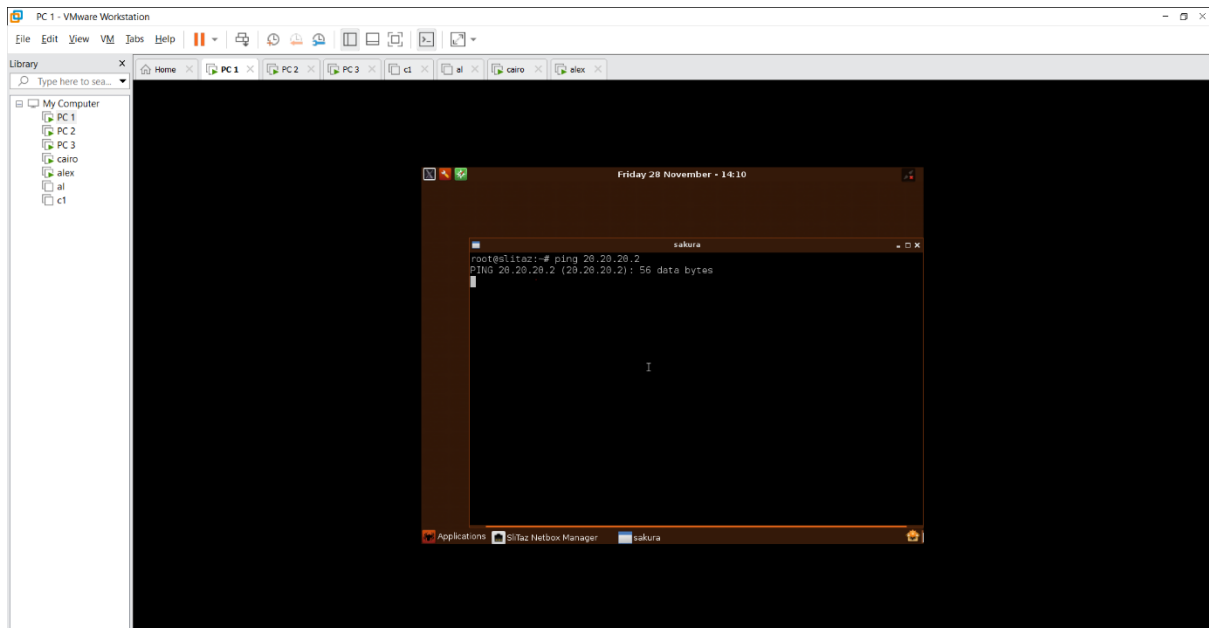
From PC-3 to PC-1



Comment:

The connectivity test results between PC-1 and PC-3 are successful because we established traffic selector from the network 10.10.10.0 to 30.30.30.0

Ping from PC-1 to PC-2



Comment:

The connectivity test results between PC-1 and PC-2 have failed because there is no traffic selector between network 10 and network 20 so the result is as expected